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Example : Pressure (P), Volume (V), Temperature(T), Internal energy (U), Enthalpy (H), free energy (G) etc.

Path functions:

A path function is a thermodynamic property of the system whose value depends on the path by which the system changes from its initial to final states.

Example: Work (*w*), Heat (*q*).

Work (w) will have different values if the process is carried out reversibly or irreversibly.

Internal Energy (U)

The internal energy is a characteristic property of a system which is denoted by the symbol U. The internal energy of a system is equal to the energy possessed by all its constituents namely atoms, ions and molecules. The total energy of all molecules in a system is equal to the sum of their translational energy (U_t), vibrational energy (U_v), rotational energy (U_r), bond energy (U_b), electronic energy (U_e) and energy due to molecular interactions (U_i).

Thus:

$$U = U_t + U_v + U_r + U_b + U_e + U_i$$

The total energy of all the molecules of the system is called internal energy. In thermodynamics one is concerned only with the change in internal energy (ΔU) rather than the absolute value of energy.

Importance of Internal energy

The internal energy possessed by a substance differentiates its physical structure. For example, the allotropes of carbon, namely, graphite C(graphite) and diamond C(diamond), differ from each other because they possess different internal energies and have different structures.

Characteristics of internal energy (U):

- The internal energy of a system is an extensive property. It depends on the amount of the substances present in the system. If the amount is doubled, the internal energy is also doubled.
- The internal energy of a system is a state function. It depends only upon the state variables (T, P, V, n) of the system. The change in internal energy does not depend on the path by which the final state is reached.
- The change in internal energy of a system is expressed as $\Delta U = U_f - U_i$
- In a cyclic process, there is no internal energy change. $\Delta U_{(cyclic)} = 0$
- If the internal energy of the system in the final state (U_f) is less than the internal energy of the system in its initial state (U_i), then ΔU would be negative.

  $$\Delta U = U_f - U_i = -ve \quad (U_f < U_i)$$
- If the internal energy of the system in the final state (U_f) is greater than the internal energy of the system in its initial state(U_i), then ΔU would be positive.

  $$\Delta U = U_f - U_i = +ve \quad (U_f > U_i)$$